

COURSE DESCRIPTION

Number	0418 - 301	Title	Algorithms Design and Analysis
Credit Hours	3 (3:0:3)	Coordinator	Dr. Fawaz Alazemi
Compulsory	Yes	Prerequisite	0418-111, 0418-201

Catalog Description

The course introduces formal techniques to support the design and analysis of algorithms, focusing on both the underlying mathematical theory and practical considerations of efficiency. Advanced algorithmic techniques such as divide and conquer, greedy, dynamic programming, backtracking, and randomized algorithmic techniques are studied. Sorting and graph algorithms are the subjects of investigation. The main objective of this course is that the students understand the mathematical concept used in describing the complexity of algorithms, select and apply algorithms, as well as to explain the trade-offs that exist between a range of algorithms that possess the same functionality. Furthermore, the course introduces complexity classes P and NP.

Textbook

Title:	Introduction to Algorithms.
Author:	T. Cormen, C. Leiserson, R. Rivest and C. Stein.
Publisher:	MIT Press, Latest Edition.

Reference

Title:	Algorithms Illuminated.
Author:	Ted Roughgarden
Publisher:	MIT Press, Latest Edition.

Major Topics Covered in the Course

	Topics	Hours
1	Introduction: definition of algorithms; design and analysis of algorithms; review of relevant concepts.	3
2	Algorithm Analysis Techniques: Time and space complexity; O , θ , Ω notations, asymptotic behavior; worst and average case behaviors; time and space tradeoffs; upper and lower bounds on complexity.	6
3	Algorithm Design Techniques: Brute force, divide and conquer, backtracking, greedy algorithms; dynamic programming.	9
4	Search and Sort Algorithms: Heap, quick, and merge sort; sorting in linear time; lower bounds of sorting; binary search trees.	9
5	Graph algorithms: Searching graphs, best first search, 2-connectivity, minimum spanning tree; shortest path algorithms; flow networks.	6
6	Distributed algorithms: Models of distributed computing, examples of distributed algorithms.	3
7	Complexity classes: Complexity classes P and NP; NP vs. P problem; NP complete problems.	3
8	Exams	3
	Total	42

Grading

20%	Midterm 1
20%	Midterm 2
20%	Coursework (homework/Quizzes/Projects)
40%	Final

Course Outcomes

Students should be able to:

1. Determine time and space complexities of simple algorithms in big O, θ , Ω notation.
2. Determine proof of correctness for simple algorithms
3. Demonstrate understanding of algorithms design techniques.
4. Demonstrate understanding of complexity classes.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	SO1	SO2	SO3	SO4	SO5	SO6
1	H					H
2	H					
3	H	H				
4	M					H

Last updated: September 2023